PMO



BUKC – Spring 2026

Project Management Office (PMO)

Lecture # 03

Functions of PMO

- **PMO functions** may include all **formal** and **informal activities** and **practices** in the **projects** in which the **PMO** is engaged **directly** or **indirectly** for the **integration of business functionality** in the organization
- The **PMO Function Models** enumerate what is **needed** in the project management environment to **effectively** conduct **project management oversight, control** or **support**
- The actual **implementation of PMO functions** in an organization depends on the **appropriate use** and **fit** within the already devised systems and operations and can be updated, as per requirement and mutual consensus

PMO Function Models

The **20 PMO functions** are grouped into **five function categories/ domains to deliberate** and **guide** development of **PMO operational capability**

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PRACTICE MANAGEMENT	INFRASTRUCTURE MANAGEMENT	RESOURCE INTEGRATION	TECHNICAL SUPPORT	BUSINESS ALIGNMENT
• PROJECT MANAGEMENT METHODOLOGY	• PROJECT GOVERNANCE	• RESOURCE MANAGEMENT	• PROJECT MENTORING	• PORTFOLIO MANAGEMENT
• PROJECT TOOLS	• ASSESSMENT	• TRAINING AND EDUCATION	• PROJECT PLANNING	• CUSTOMER RELATIONSHIP MANAGEMENT
• STANDARDS AND METRICS	• ORGANIZATION AND STRUCTURE	• CAREER DEVELOPMENT	• PROJECT AUDITING	• VENDOR RELATIONSHIP MANAGEMENT
• KNOWLEDGE MANAGEMENT	• FACILITIES AND EQUIPMENT	• TEAM DEVELOPMENT	• PROJECT RECOVERY	• BUSINESS PERFORMANCE

PMO Function Models

It is never intended that the PMO may replace organizational business functionality or functional departments. Rather, the PMO has responsibility to work in collaboration with them to facilitate adapting the business functionality in the project management environment

Function Category # 1 – **Practice Management**

*It provides a **common approach** and **frame of reference** for conducting **project management activities** within an organization to develop an **effective** organizational **project management capability***

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- This PMO function area;

a. Establishes **Project Management Methodologies**
b. Introduces associated **Project Management Tools**
c. Specifies relevant **Project Performance Standards** and **Metrics**
d. Contributes to **Project Knowledge Management**

Function Category # 1– Practice Management – a. Project Management Methodology

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- The Project Management Methodology is a set of principles and practices to guide projects for the **business integration** and to ensure **optimum performance**
- Establishes the **Standard Approach to Project Management** that should be responsive to the needs of the key project stakeholders and guide them through the key activities of project management
- Evaluation of **Project Performance Data** for analyzing status and make coarse corrections to achieve project's deliverables in time and within budget
- Once a methodology is set into place, the functional & business units **can no longer use old practices** that may misalign with or be counterproductive to the project management methodology

Function Category # 1– Practice Management – **b. Project Management Tools**

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- Determine the **types** of **project management tools** that are needed and can fit within the respective org's proj management environment
- Provide **training** in the use of proj management tools for users
- Implement and monitor the **consistent use** of proj management tools
- Ensure **integration of tools** with existing systems and processes
- Evaluate the tools performance

Function Category # 1– Practice Management – **c. Standards and Metrics**

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- Identify **accepted/ desired standards** and **metrices** to be used in the proj management environment
- **Implement** the **standards/ metrics** and manage proj's technical and business performances to the **acceptance limits/ criteria**
- Establish **consistent oversight** and **control** to ascertain the desired results and taking **corrective/ preventive actions**, where required

Function Category # 1– Practice Management – **d. Proj Knowledge Management**

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- Establish contours of **Proj Knowledge Management Framework**
- Develop/ Acquire the **Proj Knowledge Management System**
- Facilitate **collaboration** among proj managers, proj teams, and proj stakeholders to manage activities of **geographically dispersed teams**
- Capture **individuals' wisdom**, **experiences** and **lessons learned in the repository** to be utilized by all

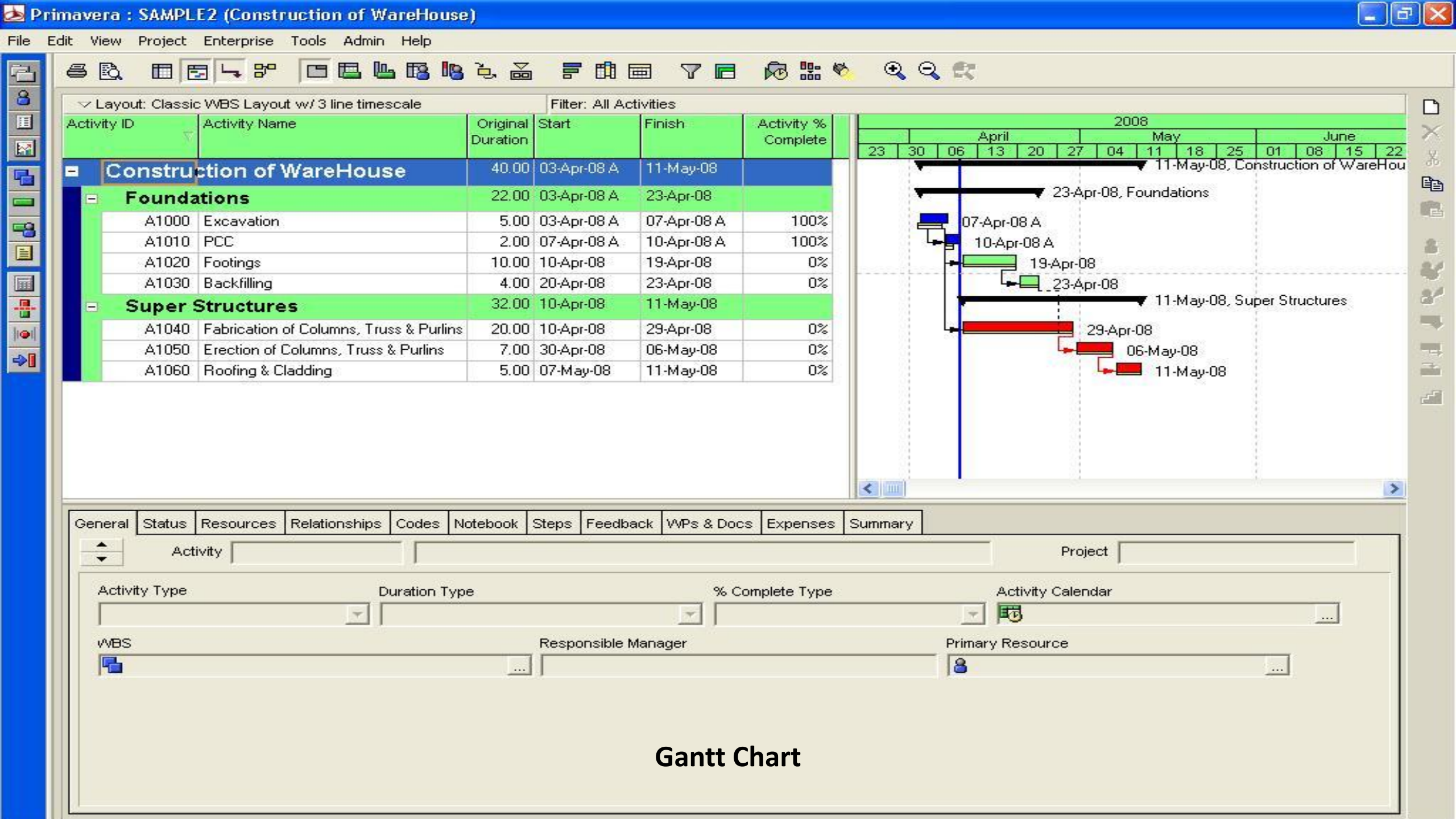
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Linked Slides

Function Category # 1– Practice Management – a. **Proj Mgmt Methodology**

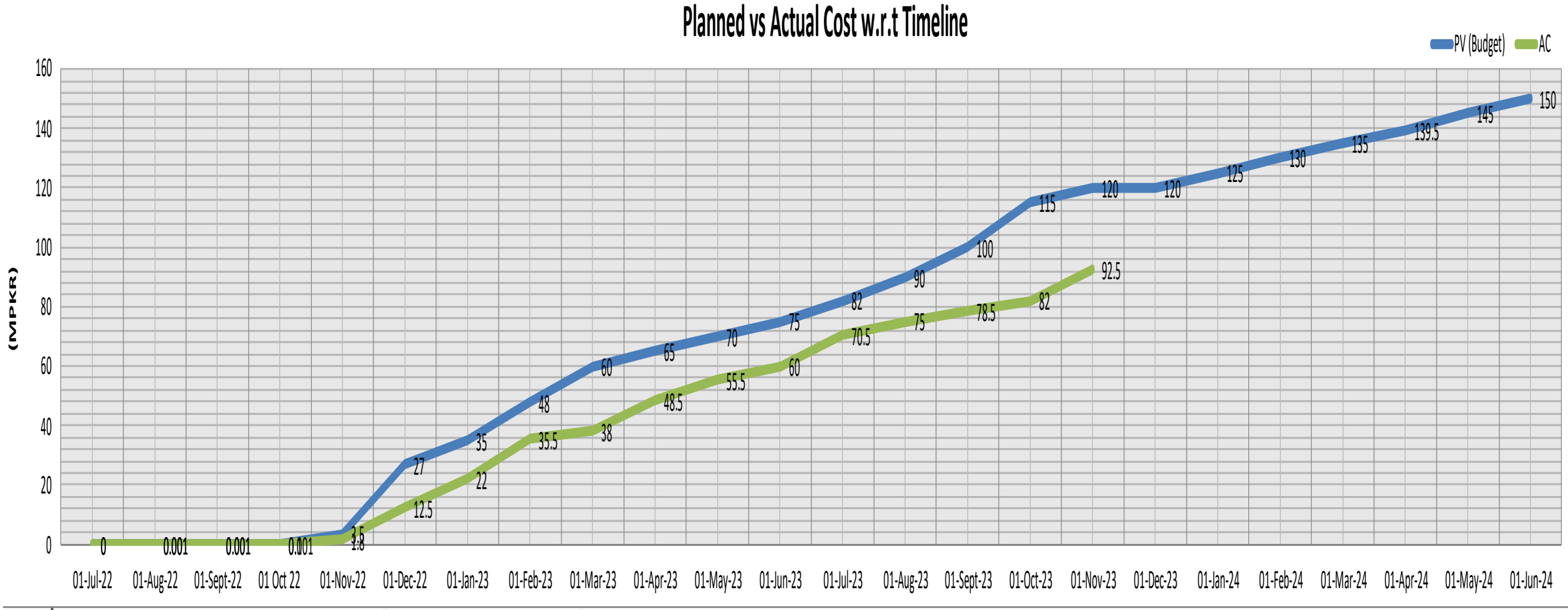
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- **Approach to Project Management**
  - Waterfall
  - Agile
  - Hybrid
- **Standardization of Methods & Processes**
  - Availability of **Project Management Manual/ Guide** detailing definition of processes, tools, calendar of activities etc
  - A centrally managed up to date **Projects' Repository** that serve as the **one window operation** for project's initiation and closure


Primavera : SAMPLE2 (Construction of WareHouse)
File Edit View Project Enterprise Tools Admin Help
Layout: Classic WBS Layout w/ 3 line timescale
Filter: All Activities
Activity ID
Activity Name
Original Duration
Start
Finish
Activity % Complete
Construction of WareHouse 40.00 03-Apr-08 A 11-May-08
Foundations 22.00 03-Apr-08 A 23-Apr-08
A1000 Excavation 5.00 03-Apr-08 A 07-Apr-08 A 100%
A1010 PCC 2.00 07-Apr-08 A 10-Apr-08 A 100%
A1020 Footings 10.00 10-Apr-08 19-Apr-08 0%
A1030 Backfilling 4.00 20-Apr-08 23-Apr-08 0%
Super Structures 32.00 10-Apr-08 11-May-08
A1040 Fabrication of Columns, Truss & Purlins 20.00 10-Apr-08 29-Apr-08 0%
A1050 Erection of Columns, Truss & Purlins 7.00 30-Apr-08 06-May-08 0%
A1060 Roofing & Cladding 5.00 07-May-08 11-May-08 0%
2008
April
May
June
23 30 06 13 20 27 04 11 18 25 01 08 15 22
11-May-08, Construction of WareHou
23-Apr-08, Foundations
07-Apr-08 A
10-Apr-08 A
19-Apr-08
23-Apr-08
11-May-08, Super Structures
29-Apr-08
06-May-08
11-May-08
General Status Resources Relationships Codes Notebook Steps Feedback WPs & Docs Expenses Summary
Activity
Project
Activity Type
Duration Type
% Complete Type
Activity Calendar
WBS
Responsible Manager
Primary Resource

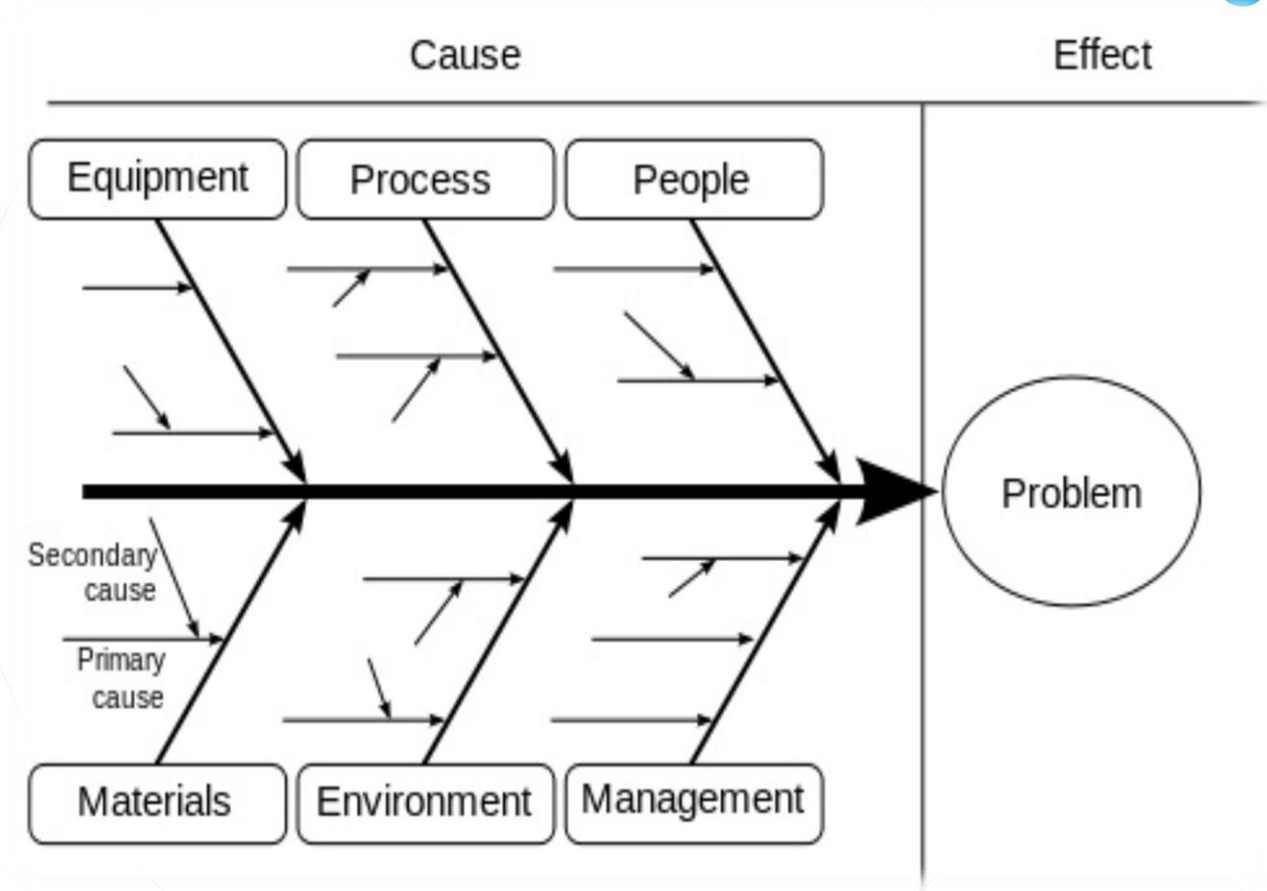
Gantt Chart

Function Category # 1– Practice Management –
b. Tools & Techniques



S Curve

Function Category # 1– Practice Management –
b. Tools & Techniques

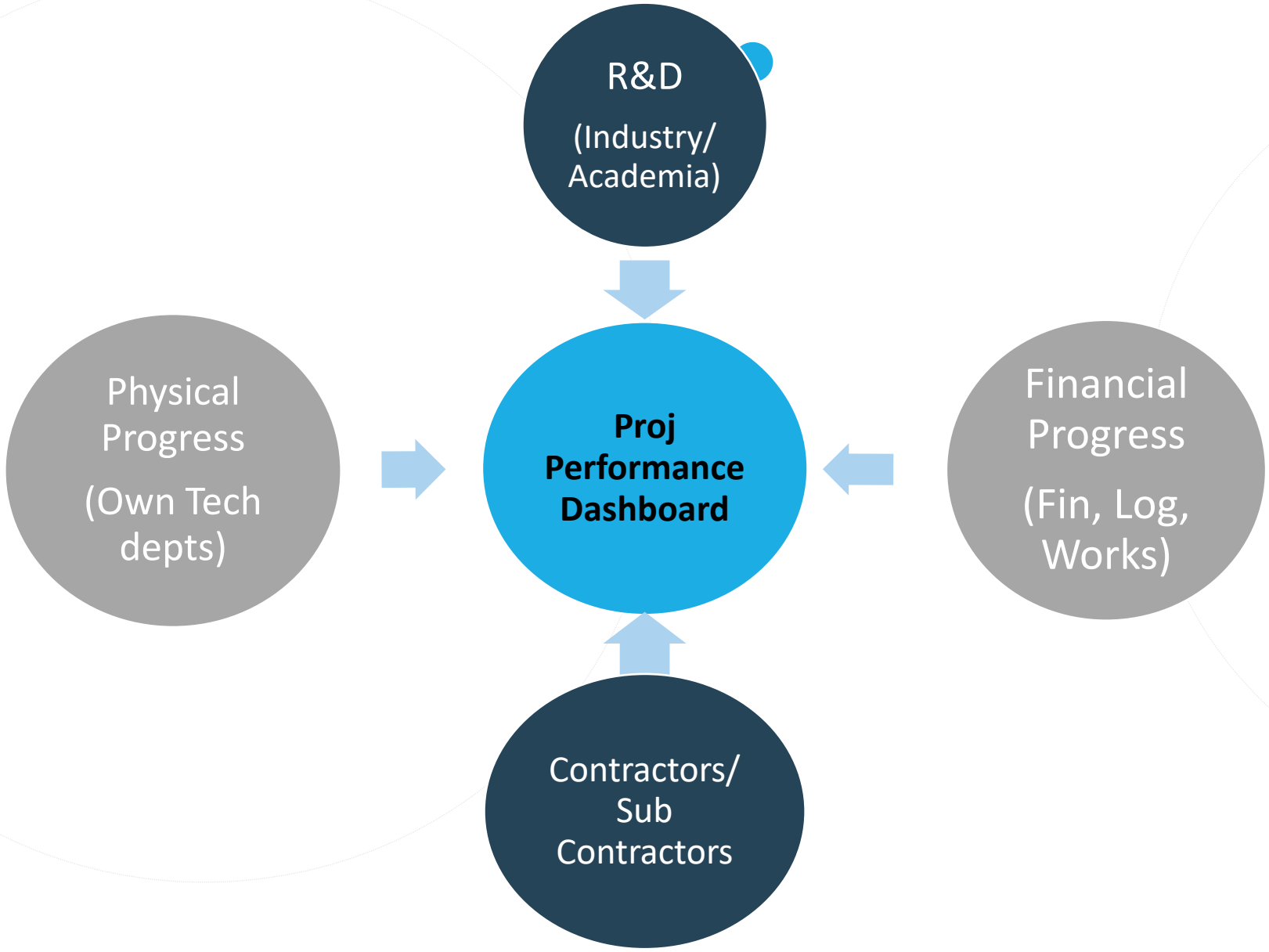


Fish bone diagram



Six Sigma

Function Category # 1– Practice Management –
b. Tools & Techniques

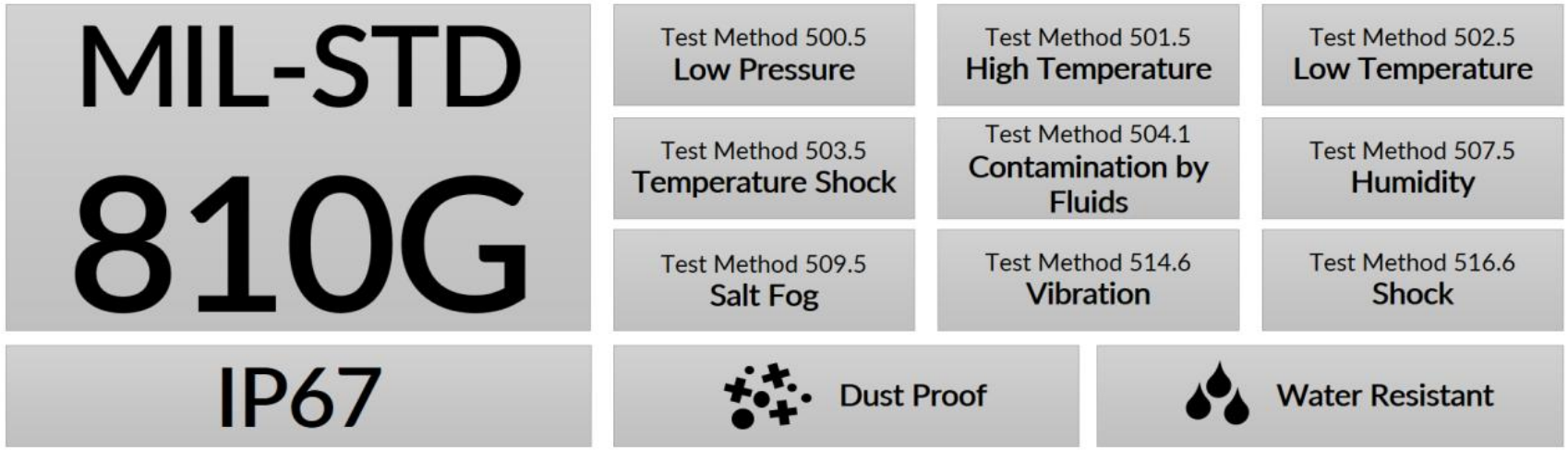


Function Category # 1– Practice Management – c. **Standards and Metrics**

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Function Category # 1– Practice Management –
d. Proj Knowledge Management

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Topics

- & Processes 2
- Accident Investigation 1
- Acquisition / procurement strategy and planning 20
- Administration / Organization 2
- Advanced planning of safety systems 1
- Business processes 4
- Center distribution of programs and management activities 3
- Communication Systems 1
- Communications between different offices and contractor personnel 4
- Configuration and data management 3
- Configuration Change Control 1
- Configuration control and data management 2
- Contractor relationships 7
- Cross Agency coordination 2
- DDT/E 2
- Early requirements and standards definition 6
- Engineering Design 3
- Engineering design and project processes and standards 7

Organization

Abstract

The lack of visibility during the design process and the inability to inspect drawings for the HIM Power Assembly (HPA), an element of the Hardware Interface Module (HIM) resulted in numerous product modifications to address both safety and reliability issues – one of which resulted in a fire in an HPA in Orbiter Processing Facility (OPF) 3. Active involvement by end item sustainers into the vendor design and production processes would have minimized these issues.

Topic(s)

Configuration control and data management, Acquisition / procurement strategy and planning, Business processes, Communications between different offices and contractor personnel, Configuration and data management, Contractor relationships, Advanced planning of safety systems, Configuration Change Control, Product Assurance, Quality, Reliability, Engineering design and project processes and standards, Long term sustainability and maintenance planning, Planning of requirements verification processes

Guidance for NASA Selection & Application of DC-DC Converters

2008-07-13

Organization

Abstract

Numerous NASA projects have suffered severe cost and schedule impacts due to problems with hybrid DC-DC converter application, quality, and reliability. Although there have been a few in-flight failures, most problems and failures have occurred during flight system development and test. This write-up summarizes DC-DC converter lessons learned that were documented in 2008 by a NASA NESC study that advises flight projects on device selection, purchase, and test.

Topic(s)

Categories, Engineering Design, Integration and Testing, Manufacturing and Assembly, Missions and Systems Requirements Definition, Program Management, Safety and Mission Assurance, Systems Engineering and Analysis, Flight Equipment, Hardware, Industrial Operations, Payloads, Risk Management / Assessment, Spacecraft, Power, Spacecraft and Spacecraft Instruments, Review boards, Acquisition / procurement strategy and planning, Early requirements and standards definition, Product Assurance, Quality, Reliability, Review systems and boards, Engineering design and project processes and standards, Long term sustainability and maintenance planning, Mission and systems trade studies